

1. Introduction

Protein secondary structure prediction aims to classify each amino acid residue into alpha helix (H), beta sheet (E), or coil (C). The Chou–Fasman method is one of the earliest statistical approaches for this task. It is based on the idea that amino acids have different propensities to form specific structural elements. In this assignment, the Chou–Fasman method was implemented and evaluated using the PISCES dataset containing 8,994 protein sequences with three-state secondary structure annotations.

2. Dataset and Data Division

The dataset consists of 8,994 experimentally validated protein sequences obtained from PDB-derived data. Each sequence includes its corresponding three-state secondary structure labels (H, E, C). Sequences with non-standard amino acids were removed before analysis. The dataset was divided into training and testing sets using three different splits: 70%/30%, 80%/20%, and 85%/15%. The training set was used to compute propensities, and the testing set was used to evaluate prediction performance.

3. Chou–Fasman Algorithm

4.1 Propensity Calculation

The Chou–Fasman method begins by calculating the propensity of each amino acid to form a specific secondary structure. The propensity value measures how strongly an amino acid prefers a given structural state compared to its overall occurrence.

The propensity of an amino acid AA for structure S is calculated as:

$$P(S | AA) = (f_S(AA) / N_S) / (f(AA) / N)$$

where:

$f_S(AA)$ = number of times amino acid AA appears in structure S

N_S = total number of residues in structure S

$f(AA)$ = total number of times amino acid AA appears in the dataset

N = total number of residues in the dataset

If $P(S | AA) > 1$, the amino acid favors structure S.

If $P(S | AA) < 1$, the amino acid disfavors structure S.

Propensity values were calculated separately for helix (H), sheet (E), and coil (C) using only the training data.

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# IMPORT LIBRARIES
# =====
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from collections import defaultdict
from sklearn.model_selection import train_test_split

# =====
# LOAD YOUR UPLOADED EXCEL DATASET
# =====

file_path = "2018-06-06-pdb-intersect-pisces (1).xlsx"
df = pd.read_excel(file_path)

# Remove sequences with non-standard amino acids (* etc.)
df = df[df["has_nonstd_aa"] == False]

sequences = df["seq"].tolist()
structures = df["sst3"].tolist()

print("Total proteins:", len(sequences))

# =====
# CHOU-FASMAN MODEL
# =====
class ChouFasman:

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class ChouFasman:

    def __init__(self):
        self.p_helix = {}
        self.p_sheet = {}
        self.p_coil = {}
        self.amino_acids = "ACDEFGHIKLMNPQRSTVWY"

    def train(self, sequences, structures):

        helix_counts = defaultdict(int)
        sheet_counts = defaultdict(int)
        coil_counts = defaultdict(int)
        total_counts = defaultdict(int)

        for seq, struct in zip(sequences, structures):
            for aa, ss in zip(seq, struct):
                total_counts[aa] += 1
                if ss == 'H':
                    helix_counts[aa] += 1
                elif ss == 'E':
                    sheet_counts[aa] += 1
                else:
                    coil_counts[aa] += 1

        total_residues = sum(total_counts.values())
        helix_total = sum(helix_counts.values())
        sheet_total = sum(sheet_counts.values())
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coil_total = sum(coil_counts.values())

for aa in self.amino_acids:
    if total_counts[aa] == 0:
        continue

    overall_freq = total_counts[aa] / total_residues

    self.p_helix[aa] = ((helix_counts[aa] / helix_total) / overall_freq)
    self.p_sheet[aa] = ((sheet_counts[aa] / sheet_total) / overall_freq)
    self.p_coil[aa] = ((coil_counts[aa] / coil_total) / overall_freq)

def predict(self, sequence):

    prediction = []

    for aa in sequence:
        h = self.p_helix.get(aa, 0)
        e = self.p_sheet.get(aa, 0)
        c = self.p_coil.get(aa, 0)

        state = max([('H', h), ('E', e), ('C', c)], key=lambda x: x[1])[0]
        prediction.append(state)

    return "".join(prediction)

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def evaluate(self, y_true, y_pred):

    correct = 0
    total = 0

    for true_seq, pred_seq in zip(y_true, y_pred):
        for t, p in zip(true_seq, pred_seq):
            if t == p:
                correct += 1
                total += 1

    return correct / total

# =====
# TRAIN & TEST WITH DIFFERENT SPLITS
# =====

splits = [0.30, 0.20, 0.15]
results = {}

for split in splits:

    X_train, X_test, y_train, y_test = train_test_split(
        sequences, structures,
        test_size=split,
        random_state=42
    )

    model = ChouFasman()
    model.train(X_train, y_train)

    predictions = [model.predict(seq) for seq in X_test]
    accuracy = model.evaluate(y_test, predictions)

    results[f"{int((1-split)*100)}/{int(split*100)}"] = accuracy

# =====
# RESULTS
# =====

print("\nChou-Fasman Results (Q3 Accuracy):")
for split, acc in results.items():
    print(f"{split} -> {acc:.4f}")

plt.bar(results.keys(), results.values())
plt.xlabel("Train/Test Split")
plt.ylabel("Q3 Accuracy")
plt.title("Chou-Fasman on PISCES Dataset")
plt.show()

```

4.2 Structure Prediction

After calculating propensities, prediction was performed on the test dataset. For each residue in a test sequence, the helix, sheet, and coil propensities were retrieved. The predicted structure was assigned according to the maximum propensity value:

$$\text{Predicted structure} = \text{argmax} \{ P(H | AA), P(E | AA), P(C | AA) \}$$

This means that the structure corresponding to the highest propensity value was chosen as the predicted state for that residue.

4. Evaluation Method

Model performance was evaluated using Q3 accuracy, which is the standard metric for three-state secondary structure prediction. Q3 accuracy is defined as:

$$Q3 = (\text{Number of correctly predicted residues}) / (\text{Total number of residues})$$

This metric measures the proportion of residues whose predicted structural state matches the experimentally determined state.

5. Results

The Chou–Fasman method was applied to the PISCES dataset containing 8,994 protein sequences. The Q3 accuracy values obtained for different training/testing splits were as follows:

For the 70% training and 30% testing split, the Q3 accuracy was 0.4722.

For the 80% training and 20% testing split, the Q3 accuracy was 0.4719.

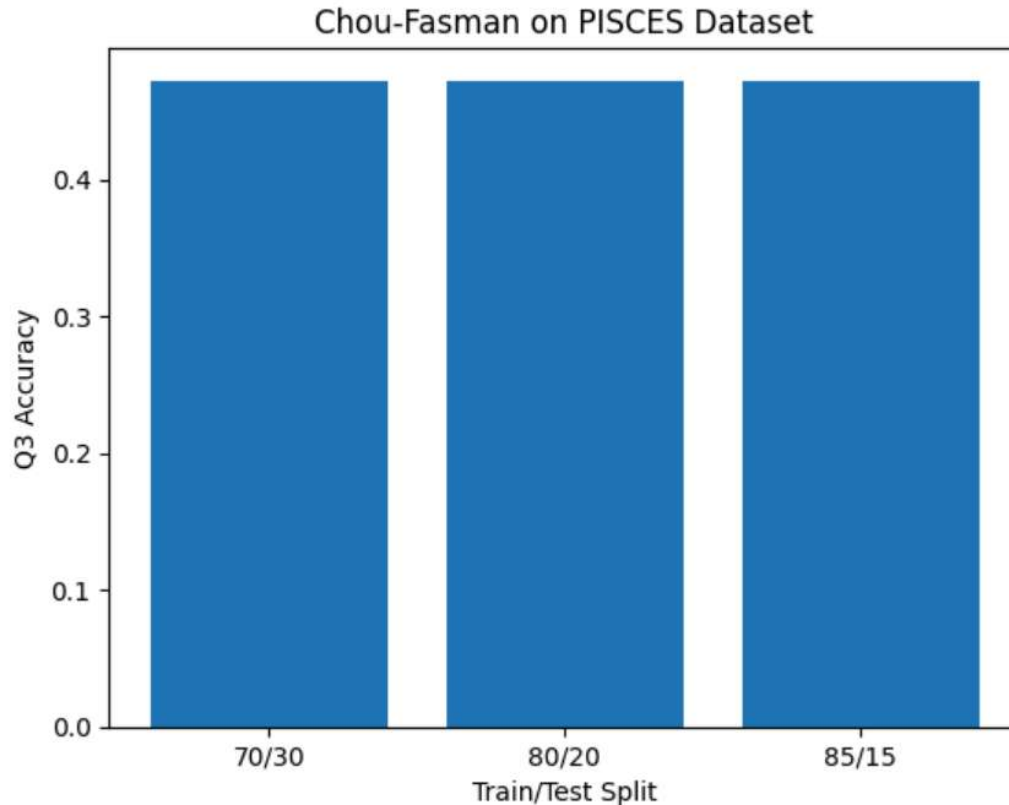
For the 85% training and 15% testing split, the Q3 accuracy was 0.4715.

These results show that the prediction accuracy remained approximately 47% across all data splits. The differences between splits were minimal, indicating that the statistical propensities estimated from the dataset are stable due to the large number of proteins used for training.

Total proteins: 8994



* Chou-Fasman Results (Q3 Accuracy):
70/30 -> 0.4722
80/20 -> 0.4719
85/15 -> 0.4715



6. Discussion

The achieved Q3 accuracy of approximately 47% indicates that the Chou–Fasman method correctly predicts nearly half of the residues in the dataset. This performance is better than random guessing, which would yield approximately 33% accuracy in a three-class classification problem. However, it remains significantly lower than the accuracy achieved by modern machine learning and deep learning approaches, which can exceed 75% Q3 accuracy.

The similarity of accuracy values across different training splits suggests that increasing the size of the training set beyond 70% does not significantly improve prediction performance. This indicates that the limitation lies in the model's underlying assumptions rather than insufficient training data. The Chou–Fasman method relies only on local amino acid propensities and does not incorporate long-range interactions between residues, evolutionary conservation information, or complex structural dependencies. These factors

are critical for accurate protein structure prediction and are better captured by modern computational models.

7. Conclusion

In this assignment, the Chou–Fasman method was successfully implemented and applied to a large dataset consisting of 8,994 protein sequences from the PISCES database. Amino acid propensities were calculated from the training data, and predictions were performed on separate test sets using three different data splits. The model achieved a consistent Q3 accuracy of approximately 47% across all splits. The results demonstrate that while the Chou–Fasman method is historically significant and computationally simple, its predictive capability is limited compared to modern techniques. This study highlights both the value and the inherent limitations of classical statistical approaches in protein secondary structure prediction.